Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-throughput cell line development and clone screening for bispecific antibodies			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	BIO · Medical		BIO · Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes/Bio-Equipment	Bio-process Technologies	Medical Bio	Pharmaceutical Bio-based Technologies and Systems
Concept of Proposed Technology	Bispecific antibodies represent a high-value therapeutic modality but present substantial manufacturing challenges due to low expression titers and suboptimal product quality attributes, including inefficient heterodimerization[1]. The proposed technology introduces a **data-driven, high-throughput cell line development (CLD) and process development (PD) platform** based on **heritable DNA barcoding and lineage tracing**. Each candidate clone is uniquely labeled with a DNA barcode that can be linked to multiple performance attributes, including vector design, growth characteristics, productivity, gene expression profiles, and product quality parameters. This approach enables the screening of **more than 1,000-fold greater numbers of clones** in scaled-up bioreactor environments compared with existing technologies. The resulting large-scale, multi-omic datasets provide a foundation for **AI-enabled applications** and support accelerated selection of optimal production clones.			

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	By integrating clone screening with biomanufacturing-relevant process conditions, the platform aims to reduce development timelines from vector design to regulatory filing and improve the robustness of bispecific antibody production workflows.
Rationale for Joint R&D Project	1. **Necessity of Industrial Scale-Down Models:** To achieve commercial validation, our platform requires testing in industrial-grade environments. We aim to collaborate with a Korean partner to validate our platform on established scale-down models (e.g., 2L bioreactors or Ambr 250 systems) and pilot-scale bioreactors. This cooperative approach allows us to prove that the best clones identified by our approach maintain their high performance when subjected to the industrial-scale manufacturing conditions. 2. **Access to Industry-Relevant Bispecific Candidates:** Bispecific antibodies are high-value but notoriously difficult to express due to assembly challenges. This joint R&D project offers a unique opportunity to apply our technology to a partner's pipeline of "hard-to-express" bispecific antibody candidates. By targeting specific, proprietary molecules that currently suffer from low titers or quality issues, we can demonstrate a direct improvement in manufacturing bottlenecks, effectively moving the technology to TRL 6. 3. **Unifying Cell Line and Process Development:** Current biomanufacturing often treats Cell Line Development (CLD) and Process Development (PD) as isolated stages, leading to inefficiencies when selected clones fail in downstream processing. This joint R&D aims to bridge these fragmented workflows by integrating our high-throughput barcoding data directly with a partner's bioprocess capabilities. By linking upstream clonal data to downstream performance metrics, we can select clones that are fundamentally optimized for the entire manufacturing lifecycle, thereby accelerating the combined CLD and PD timeline.
Key Words	1. High-throughput 2. Cell line development 3. Bioinformatics 4. Lineage tracing 5. Scale-up

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	To establish a high-throughput, integrated biomanufacturing platform capable of reducing the timeline from vector design to regulatory filing to less than 26 weeks. **Specific Objectives:** • Adapt a genome-wide safe harbor identification technology to Chinese Hamster Ovary (CHO) cells by screening at least 10,000 loci for stable integration sites supporting high-titer expression • Expand multiplexed clonal tracking methods to address heterodimerization and quality challenges of bispecific antibodies by screening at least 1,000 vector designs • Validate selected clones under industrial scale-down conditions using pilot-scale bioreactors in collaboration with a Korean SME partner			
R&D Target Specifications	• Identification and screening of ≥ 10,000 genomic safe harbor loci • High-throughput screening of ≥ 10,000 clones representing ≥ 1,000 vector designs in biomanufacturing-relevant conditions • Validation of approximately 12 top-performing clones in scale-down bioreactor models • Demonstrated maintenance of productivity and quality attributes at pilot scale • Reduction of combined CLD and PD timeline to less than 26 weeks			
R&D Plan	**Approach Overview:** Two complementary strategies are applied: genome-wide screening for desirable engineering interventions and single-cell screening for top-performing clones. **Phase 1: Platform Adaptation and Library Construction (Year 1)** • Adapt genome-wide safe harbor screening technology to CHO cells • Optimize DNA barcode cassette design and delivery methods • Construct a large combinatorial library of barcoded vectors for a model bispecific antibody (≥ 1,000 designs) *Deliverables:* • CHO-adapted safe harbor screening system • Barcoded bispecific antibody vector library **Phase 2: High-Throughput Screening and Candidate Selection (Year 2)** • Identify ≥ 10,000 potential safe harbor loci in the CHO genome • Generate a barcoded pool of ≥ 10,000 clones • Screen for growth, productivity, heterodimerization efficiency, and quality attributes • Isolate top-performing lineages using retrieval methods linked to barcode identity *Deliverables:* • Ranked candidate clone set • Multi-omic screening dataset • Correlation analysis of clone performance and quality attributes			

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	Phase 3: Industrial Scale-Down Validation (Year 3) • Evaluate top ~12 clones in industrial scale-down bioreactor models • Assess maintenance of titer and product quality under pilot-scale conditions • Validate predictive accuracy of the platform *Deliverables:* • Scale-down validation report • Final candidate clone set • Technical verification summary

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Develop and refine lineage tracing, DNA barcoding, and high-throughput screening methodologies • Perform genome-wide safe harbor identification and vector library construction • Conduct multi-omic data generation and analysis • Lead scientific validation and documentation of platform performance • Provide technical expertise in cell engineering and biomanufacturing research
Expected R&R of Korean SME Partner	• Provide access to industrial scale-down and pilot-scale bioreactor systems • Supply or advise on selection of difficult-to-express bispecific antibody candidates • Support downstream process verification and purification strategy development • Define analytical standards for critical quality attributes • Provide guidance on regulatory compliance and documentation aligned with IND/BLA preparation • Participate in performance evaluation and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • Identification of ≥ 10,000 genomic safe harbor loci and validation of top candidate sites • Screening of ≥ 1,000 vector designs across ≥ 10,000 clones in biomanufacturing conditions • Demonstration of reduced development timelines to less than 26 weeks • Generation of high-value intellectual property and patentable innovation **Impact** • Establishment of a next-generation high-throughput CLD and PD platform for bispecific antibodies • Improved productivity and quality of difficult-to-express biologics • Strengthening of Korean SME competitiveness in advanced biomanufacturing technologies • Acceleration of regulatory filing timelines and reduction of development costs • Contribution to national Advanced Bio strategic technology objectives

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6. References

(1) Ingavat, N., Wang, X., Kok, Y. J., Dzulkiflie, N., Loh, H. P., Leong, E., ... & Zhang, W. (2025). Affinity resin selection for efficient capture of bispecific antibodies as guided by domain composition. *Process Biochemistry,* 154, 1-11.